

FagusED · Mathematics
01 · Number

What is the key method or rule for number structure?

MAT-0001

PREGUNTA

FagusED · Mathematics
01 · Number

What is the key method or rule for sets and regions?

MAT-0002

PREGUNTA

FagusED · Mathematics
01 · Number

Which is prime?
A) 1
B) 21
C) 29
D) 39

MAT-0003

PREGUNTA

FagusED · Mathematics
01 · Number

What is HCF(18,30)?
A) 3
B) 6
C) 9
D) 90

MAT-0004

PREGUNTA

FagusED · Mathematics
01 · Number

What is 15% of 80?
A) 8
B) 12
C) 15
D) 20

MAT-0005

PREGUNTA

FagusED · Mathematics
01 · Number

Write 0.00056 in standard form.
A) 5.6×10^{-4}
B) 5.6×10^4
C) 56×10^{-4}
D) 0.56×10^{-3}

MAT-0006

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

What is the key method or rule for expressions and factorisation?

MAT-0007

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

What is the key method or rule for solving and checking equations?

MAT-0008

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

Simplify $3x + 2x - 4$.
A) $5x - 4$
B) $6x - 4$
C) $x - 4$
D) $5x + 4$

MAT-0009

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

Solve $4x - 7 = 9$.
A) 2
B) 4
C) 8
D) 16

MAT-0010

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

Factorise $x^2 - 9$.
A) $(x - 9)(x + 1)$
B) $(x - 3)^2$
C) $(x - 3)(x + 3)$
D) $x(x - 9)$

MAT-0011

PREGUNTA

FagusED · Mathematics
02 · Algebra and graphs

What is the n th term of 2, 5, 8, ...?
A) $2n + 1$
B) $3n - 1$
C) $3n + 1$
D) $n + 2$

MAT-0012

PREGUNTA

FagusED · Mathematics
03 · Coordinate geometry

What is the key method or rule for gradient from two points?

MAT-0013

PREGUNTA

FagusED · Mathematics
03 · Coordinate geometry

What is the key method or rule for midpoint and distance?

MAT-0014

PREGUNTA

FagusED · Mathematics
03 · Coordinate geometry

Gradient from (0,1) to (2,5)?
A) 1
B) 2
C) 3
D) 4

MAT-0015

PREGUNTA

FagusED · Mathematics
01 · Number

C) 29

MAT-0003

RESPUESTA

FagusED · Mathematics
01 · Number

A universal set contains all objects under discussion. $A \cap B$ is the overlap, $A \cup B$ includes either set, and A' is the complement relative to the universal set. Fill the overlap first, then the exclusive parts, then the outside region.

MAT-0002

RESPUESTA

FagusED · Mathematics
01 · Number

A prime number has exactly two positive factors. Prime factorisation writes an integer as a product of primes; the highest common factor uses the lowest common powers and the lowest common multiple uses the highest. A rational number can be written as a fraction of integers; irrational numbers cannot.

MAT-0001

RESPUESTA

FagusED · Mathematics
01 · Number

A) 5.6×10^{-4}

MAT-0006

RESPUESTA

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01 · Number

B) 12

MAT-0005

RESPUESTA

FagusED · Mathematics
01 · Number

B) 6

MAT-0004

RESPUESTA

FagusED · Mathematics
02 · Algebra and graphs

A) $5x - 4$

MAT-0009

RESPUESTA

FagusED · Mathematics
02 · Algebra and graphs

Use inverse operations on both sides and show the line where the variable is isolated. In a simultaneous system, eliminate one variable or substitute an expression for it. A quadratic may factorise; check both roots by substitution. For an inequality, reverse the sign only after multiplying or dividing by a negative number.

MAT-0008

RESPUESTA

FagusED · Mathematics
02 · Algebra and graphs

Collect like terms before expanding. Multiply every term in a bracket; factorisation reverses expansion. For $x^2 + bx + c$, find two numbers with sum b and product c . Algebraic fractions follow the same rules as numerical fractions, with excluded values where a denominator is zero.

MAT-0007

RESPUESTA

FagusED · Mathematics
02 · Algebra and graphs

B) $3n - 1$

MAT-0012

RESPUESTA

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02 · Algebra and graphs

C) $(x - 3)(x + 3)$

MAT-0011

RESPUESTA

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02 · Algebra and graphs

B) 4

MAT-0010

RESPUESTA

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03 · Coordinate geometry

B) 2

MAT-0015

RESPUESTA

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03 · Coordinate geometry

The midpoint averages the two x -values and the two y -values separately. The distance formula comes from Pythagoras: $d^2 = (\Delta x)^2 + (\Delta y)^2$. A sketch helps detect an impossible negative distance or midpoint outside the segment.

MAT-0014

RESPUESTA

FagusED · Mathematics
03 · Coordinate geometry

The gradient of the line from (x_1, y_1) to (x_2, y_2) is change in y divided by change in x . A positive gradient rises to the right and a negative gradient falls. A horizontal line has gradient zero; a vertical line has undefined gradient.

MAT-0013

RESPUESTA

FagusED · Mathematics
03 · Coordinate geometry

Midpoint of (2,4) and (8,10)?
A) (6,14)
B) (5,7)
C) (3,3)
D) (10,14)

MAT-0016

PREGUNTA

FagusED · Mathematics
03 · Coordinate geometry

Distance between (0,0) and (3,4)?
A) 4
B) 5
C) 6
D) 7

MAT-0017

PREGUNTA

FagusED · Mathematics
03 · Coordinate geometry

Which line is parallel to $y = 3x + 2$?
A) $y = -3x + 4$
B) $y = x/3 + 4$
C) $y = 3x - 7$
D) $y = -x/3 + 1$

MAT-0018

PREGUNTA

FagusED · Mathematics
04 · Geometry

What is the key method or rule for angles and polygons?

MAT-0019

PREGUNTA

FagusED · Mathematics
04 · Geometry

What is the key method or rule for congruence and similarity?

MAT-0020

PREGUNTA

FagusED · Mathematics
04 · Geometry

Sum of angles in a triangle?
A) 90°
B) 180°
C) 270°
D) 360°

MAT-0021

PREGUNTA

FagusED · Mathematics
04 · Geometry

Interior angle sum of a hexagon?
A) 360°
B) 540°
C) 720°
D) 900°

MAT-0022

PREGUNTA

FagusED · Mathematics
04 · Geometry

One exterior angle of a regular decagon?
A) 18°
B) 36°
C) 72°
D) 144°

MAT-0023

PREGUNTA

FagusED · Mathematics
04 · Geometry

If lengths scale by 3, areas scale by
A) 3
B) 6
C) 9
D) 27

MAT-0024

PREGUNTA

FagusED · Mathematics
05 · Mensuration

What is the key method or rule for units and area?

MAT-0025

PREGUNTA

FagusED · Mathematics
05 · Mensuration

What is the key method or rule for circles, arcs and sectors?

MAT-0026

PREGUNTA

FagusED · Mathematics
05 · Mensuration

1 m^2 equals how many cm^2 ?
A) 100
B) 1000
C) 10000
D) 1000000

MAT-0027

PREGUNTA

FagusED · Mathematics
05 · Mensuration

Area of a triangle with base 8 cm and height 5 cm?
A) 13
B) 20
C) 40
D) 80

MAT-0028

PREGUNTA

FagusED · Mathematics
05 · Mensuration

Circumference of a circle with radius 4?
A) 4π
B) 8π
C) 16π
D) 32π

MAT-0029

PREGUNTA

FagusED · Mathematics
05 · Mensuration

Volume of a cylinder with radius 2 and height 5?
A) 10π
B) 20π
C) 25π
D) 40π

MAT-0030

PREGUNTA

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03 · Coordinate geometry

C) $y = 3x - 7$

MAT-0018

RESPUESTA

FagusED · Mathematics
03 · Coordinate geometry

B) 5

MAT-0017

RESPUESTA

FagusED · Mathematics
03 · Coordinate geometry

B) (5,7)

MAT-0016

RESPUESTA

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04 · Geometry

B) 180°

MAT-0021

RESPUESTA

FagusED · Mathematics
04 · Geometry

Congruent triangles match exactly in side lengths and angles, often by SSS, SAS, ASA or RHS. Similar figures have equal corresponding angles and lengths in a common ratio k . Corresponding areas change by k^2 and volumes by k^3 . Match corresponding vertices before calculating.

MAT-0020

RESPUESTA

FagusED · Mathematics
04 · Geometry

Angles on a straight line total 180° , around a point 360° and inside a triangle 180° . Parallel lines yield corresponding and alternate equal angles and co-interior supplementary angles. A polygon with n sides has interior angle sum $(n - 2) \times 180^\circ$.

MAT-0019

RESPUESTA

FagusED · Mathematics
04 · Geometry

C) 9

MAT-0024

RESPUESTA

FagusED · Mathematics
04 · Geometry

B) 36°

MAT-0023

RESPUESTA

FagusED · Mathematics
04 · Geometry

C) 720°

MAT-0022

RESPUESTA

FagusED · Mathematics
05 · Mensuration

C) 10000

MAT-0027

RESPUESTA

FagusED · Mathematics
05 · Mensuration

The circle circumference is $2\pi r$ and area is πr^2 . A sector with central angle θ takes fraction $\theta/360$ of a full circle, so its arc length is $(\theta/360)2\pi r$ and area is $(\theta/360)\pi r^2$. Include straight radii when asked for sector perimeter.

MAT-0026

RESPUESTA

FagusED · Mathematics
05 · Mensuration

Convert lengths before calculating. Area is measured in square units: $1 \text{ m}^2 = 10000 \text{ cm}^2$; volume is in cubic units: $1 \text{ m}^3 = 1000000 \text{ cm}^3$. A triangle has area $1/2 \times \text{base} \times \text{perpendicular height}$, and a trapezium has area $1/2 \times \text{sum of parallel sides} \times \text{height}$.

MAT-0025

RESPUESTA

FagusED · Mathematics
05 · Mensuration

B) 20π

MAT-0030

RESPUESTA

FagusED · Mathematics
05 · Mensuration

B) 8π

MAT-0029

RESPUESTA

FagusED · Mathematics
05 · Mensuration

B) 20

MAT-0028

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

What is the key method or rule for right triangles?

MAT-0031

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

What is the key method or rule for sine, cosine and tangent?

MAT-0032

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

Hypotenuse of sides 5 and 12?
A) 11
B) 12
C) 13
D) 17

MAT-0033

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

For opposite 3, hypotenuse 5, $\sin \theta$ is
A) $\frac{3}{5}$
B) $\frac{4}{5}$
C) $\frac{5}{3}$
D) $\frac{3}{4}$

MAT-0034

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

For adjacent 4, opposite 3, $\tan \theta$ is
A) $\frac{4}{3}$
B) $\frac{3}{4}$
C) $\frac{3}{5}$
D) $\frac{4}{5}$

MAT-0035

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

Which is the sine rule?
A) $a/\sin A = b/\sin B$
B) $a^2 + b^2 = c^2$ for all triangles
C) $\sin A + \sin B = 1$
D) $a \cos A = b \cos B$

MAT-0036

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

What is the key method or rule for moving a shape?

MAT-0037

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

What is the key method or rule for enlargement and scale?

MAT-0038

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

Translate (2,3) by (4,-1).
A) (6,2)
B) (-2,4)
C) (8,3)
D) (2,-1)

MAT-0039

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

Reflect (3,-2) in the x-axis.
A) (-3,-2)
B) (-3,2)
C) (3,2)
D) (2,-3)

MAT-0040

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

Rotate (2,1) 180° about origin.
A) (1,2)
B) (-2,-1)
C) (2,-1)
D) (-1,2)

MAT-0041

PREGUNTA

FagusED · Mathematics
07 · Transformations and vectors

Enlarge (1,-3) by scale factor 2 about origin.
A) (2,-6)
B) (-2,6)
C) (1,-6)
D) (3,-1)

MAT-0042

PREGUNTA

FagusED · Mathematics
08 · Probability

What is the key method or rule for single events?

MAT-0043

PREGUNTA

FagusED · Mathematics
08 · Probability

What is the key method or rule for sample space and combinations?

MAT-0044

PREGUNTA

FagusED · Mathematics
08 · Probability

$P(A) = 0.35$. What is $P(\text{not } A)$?
A) 0.35
B) 0.65
C) 1.35
D) 0.75

MAT-0045

PREGUNTA

FagusED · Mathematics
06 · Trigonometry

C) 13

MAT-0033

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

For a marked acute angle in a right triangle, sine = opposite/hypotenuse, cosine = adjacent/hypotenuse, tangent = opposite/adjacent. Label sides relative to that angle before choosing. Use inverse trig for an angle and check the calculator is in degrees.

MAT-0032

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

Pythagoras gives $\text{hypotenuse}^2 = \text{other side}^2 + \text{other side}^2$. The hypotenuse is opposite the right angle. For a missing shorter side, subtract squared lengths before taking a square root. In 3D, use one right triangle to find a face diagonal, then another for a space diagonal.

MAT-0031

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

A) $a/\sin A = b/\sin B$

MAT-0036

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

B) 3/4

MAT-0035

RESPUESTA

FagusED · Mathematics
06 · Trigonometry

A) 3/5

MAT-0034

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

A) (6,2)

MAT-0039

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

An enlargement with scale factor k sends a point X from centre O to X' with $OX' = k OX$ as directed. Negative scale factors place images across the centre. Lengths change by $|k|$ and areas by k^2 . Use lines through corresponding vertices to locate the centre.

MAT-0038

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

A translation moves each point by the same column vector, such as (3,-2): right 3 and down 2. A reflection uses a mirror line and reverses orientation. A rotation requires its centre, angle and direction. Give a transformation in full to distinguish motions with similar images.

MAT-0037

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

A) (2,-6)

MAT-0042

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

B) (-2,-1)

MAT-0041

RESPUESTA

FagusED · Mathematics
07 · Transformations and vectors

C) (3,2)

MAT-0040

RESPUESTA

FagusED · Mathematics
08 · Probability

B) 0.65

MAT-0045

RESPUESTA

FagusED · Mathematics
08 · Probability

List the full sample space systematically using a table or tree. For either event A or event B , $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If events cannot occur together, the overlap is zero. With equally likely pair outcomes, count ordered pairs rather than just distinct totals.

MAT-0044

RESPUESTA

FagusED · Mathematics
08 · Probability

A probability lies between 0 and 1. For equally likely outcomes, $P(A) = \text{favourable outcomes} / \text{total outcomes}$. Complement $P(\text{not } A) = 1 - P(A)$. Experimental relative frequency approximates probability when trials are numerous, but is not guaranteed to match exactly.

MAT-0043

RESPUESTA

FagusED · Mathematics
08 · Probability

P(heads twice) for a fair coin?
A) $\frac{1}{2}$
B) $\frac{1}{4}$
C) $\frac{1}{3}$
D) $\frac{1}{8}$

MAT-0046

PREGUNTA

FagusED · Mathematics
08 · Probability

Probability of a 6 on a fair die?
A) $\frac{1}{3}$
B) $\frac{1}{6}$
C) $\frac{1}{4}$
D) 6

MAT-0047

PREGUNTA

FagusED · Mathematics
08 · Probability

A certain event has probability
A) 0
B) 0.5
C) 1
D) 6

MAT-0048

PREGUNTA

FagusED · Mathematics
09 · Statistics

What is the key method or rule for
kinds of data and displays?

MAT-0049

PREGUNTA

FagusED · Mathematics
09 · Statistics

What is the key method or rule for
mean, median and spread?

MAT-0050

PREGUNTA

FagusED · Mathematics
09 · Statistics

Median of 2,4,9?
A) 2
B) 4
C) 5
D) 9

MAT-0051

PREGUNTA

FagusED · Mathematics
09 · Statistics

Mean of 3,5,7,9?
A) 5
B) 6
C) 7
D) 8

MAT-0052

PREGUNTA

FagusED · Mathematics
09 · Statistics

Range of 2,5,11,14?
A) 9
B) 11
C) 12
D) 14

MAT-0053

PREGUNTA

FagusED · Mathematics
09 · Statistics

For grouped data, the mean uses
A) lower bounds only
B) class midpoints
C) upper bounds only
D) modal class only

MAT-0054

PREGUNTA

FagusED · Mathematics
08 · Probability

C) 1

MAT-0048

RESPUESTA

FagusED · Mathematics
08 · Probability

B) $\frac{1}{6}$

MAT-0047

RESPUESTA

FagusED · Mathematics
08 · Probability

B) $\frac{1}{4}$

MAT-0046

RESPUESTA

FagusED · Mathematics
09 · Statistics

B) 4

MAT-0051

RESPUESTA

FagusED · Mathematics
09 · Statistics

Mean = sum of values/number of values. Median is the middle ordered observation; mode is the most frequent. Range = largest - smallest. Quartiles split ordered data, and interquartile range = $Q3 - Q1$. Compare typical value and spread together.

MAT-0050

RESPUESTA

FagusED · Mathematics
09 · Statistics

Categorical data name groups; numerical data are discrete counts or continuous measurements. Choose a bar chart for categories, a histogram for grouped continuous values and a scatter diagram for paired measurements. An unequal interval width in a histogram requires frequency density, not bar height equal to frequency.

MAT-0049

RESPUESTA

FagusED · Mathematics
09 · Statistics

B) class midpoints

MAT-0054

RESPUESTA

FagusED · Mathematics
09 · Statistics

C) 12

MAT-0053

RESPUESTA

FagusED · Mathematics
09 · Statistics

B) 6

MAT-0052

RESPUESTA